Examples: Soft iron and Mumetal (Nickel Iron alloy).

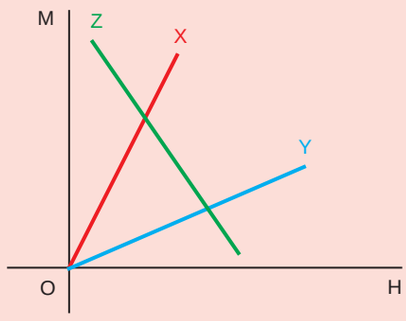
iii) Core of the transformer:

The materials with high initial permeability, large magnetic induction and thin hysteresis loop with smaller area are needed to design transformer cores.

Examples: Soft iron

EXAMPLE 3.13

The following figure shows the variation of intensity of magnetisation with the applied magnetic field intensity for three magnetic materials X, Y and Z. Identify the materials X,Y and Z.



Solution

The slope of M-H graph measures the magnetic susceptibility, which is

$$\chi_m = \frac{M}{H}$$

Material X: Slope is positive and larger value. So, it is a ferromagnetic material.

Material Y: Slope is positive and lesser value than X. So, it could be a paramagnetic material.

Material Z: Slope is negative and hence, it is a diamagnetic material.

3.3 MAGNETIC EFFECTS OF CURRENT

3.7.1 Oersted experiment

In 1820, Hans Christian Oersted while preparing for his lecture in physics noticed that electric current passing through a wire deflects the nearby magnetic compass. By proper investigation, he observed that the deflection of magnetic compass is due to the change in magnetic field produced around current carrying conductor (Figure 3.32). When the direction of current is reversed, the magnetic compass deflects in opposite direction. This lead to the development of the theory 'electromagnetism' which unifies the two branches in physics, namely electricity and magnetism.

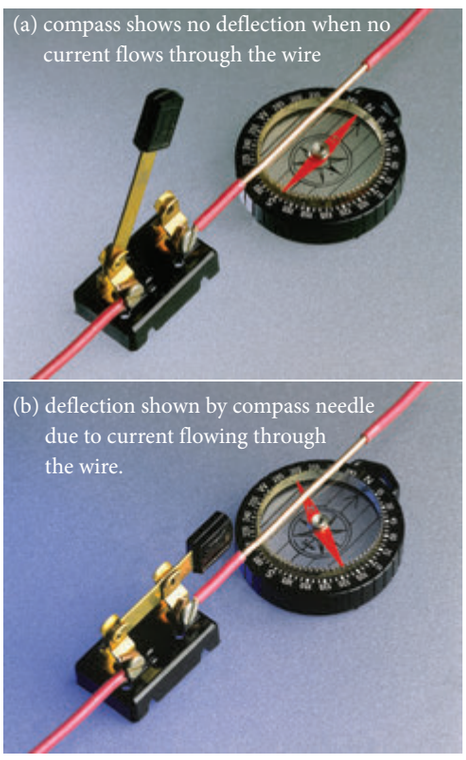


Figure 3.32 Oersted's experiment - current carrying wire and deflection of magnetic needle

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